TRIGONOMETRIC EQUATIONS

General and restricted solutions, side by side

The same solutions, written for different questions.

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COMPARED

	GENERAL SOLUTION	RESTRICTED SOLUTION
WHAT IT LISTS	every solution, in one expression with an integer n	only the solutions inside a given interval
EXAMPLE: SINE EQUALS ONE HALF	$x = \frac{\pi}{6} + 2n\pi$ or $x = \frac{5\pi}{6} + 2n\pi$	on $[0, 2\pi)$: $x = \frac{\pi}{6}$ and $x = \frac{5\pi}{6}$
ASKED FOR WHEN	the problem says "solve" or "find all solutions"	an interval is given

Sine and cosine give two solutions per period 2π when $|a| < 1$; tangent gives one per period π.